Recall that, for a given method $\mathcal{G}_h$ the definition of the adjoint method $\mathcal{G}_h^*$ is given by

$$\mathcal{G}_h^* = \mathcal{G}_{-h}^{-1}.$$

For Symplectic Euler with constraints, we view $\mathcal{G}_h$ as a mapping of the co-tangent bundle and write $(\boldsymbol{Q}, \boldsymbol{P}) := \mathcal{G}_h(\boldsymbol{q}, \boldsymbol{p})$ where

$$\begin{aligned}
\hat{\boldsymbol{P}} &= \boldsymbol{p} + h\boldsymbol{F}(\boldsymbol{q}) - hg'(\boldsymbol{q})^T\boldsymbol{\lambda}, \\
\boldsymbol{Q} &= \boldsymbol{q} + h\boldsymbol{M}^{-1}\hat{\boldsymbol{P}}, \\
\boldsymbol{P} &= \hat{\boldsymbol{P}} - h\boldsymbol{g}'(\boldsymbol{Q})^T\boldsymbol{\mu},
\end{aligned}$$

and the multipliers $\boldsymbol{\lambda}, \boldsymbol{\mu}$ are defined by the conditions $\boldsymbol{g}(\boldsymbol{Q}) = 0, \boldsymbol{g}'(\boldsymbol{Q})\boldsymbol{M}^{-1}\boldsymbol{P} = 0$.

Now replace $h$ by $-h$ and then swap $\boldsymbol{Q}$ with $\boldsymbol{q}$ and $\boldsymbol{P}$ with $\boldsymbol{p}$, resulting in

$$\begin{aligned}
\hat{\boldsymbol{P}} &= \boldsymbol{p} - hg'(\boldsymbol{q})^T\boldsymbol{\lambda}, \\
\boldsymbol{Q} &= \boldsymbol{q} + h\boldsymbol{M}^{-1}\hat{\boldsymbol{P}}, \\
\boldsymbol{P} &= \hat{\boldsymbol{P}} + h\boldsymbol{F}(\boldsymbol{Q}) - h\boldsymbol{g}'(\boldsymbol{Q})^T\boldsymbol{\mu},
\end{aligned}$$

where $\boldsymbol{\lambda}$, $\boldsymbol{\mu}$ are again defined by the constraint conditions. This defines the adjoint method, another first order method for the constrained system. By composing the maps $\mathcal{G}_{h/2}$ and $\mathcal{G}_{h/2}^*$ we get a second order symplectic method. In the sense that it is defined by first order maps which reduce to the Symplectic Euler method and its adjoint, this is an analogous scheme to the Störmer-Verlet method. Interestingly, this method is not equivalent to SHAKE or RATTLE.

Yet this is not the only constrained composition method we could have designed in analogy with Störmer-Verlet. Recall that Störmer-Verlet is also the symmetric composition of the flow maps of the two Hamiltonians $K = \boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}/2$ and $U = U(\boldsymbol{q})$. It would be natural to consider a composition of steps resolving the constrained flows on $K$ and $U$ separately. For the potential energy term, an exact solution is given by

$$\begin{aligned}
\boldsymbol{Q} &= \boldsymbol{q}, \\
\boldsymbol{P} &= \boldsymbol{p} - h\boldsymbol{\Pi}(\boldsymbol{q})\nabla U(\boldsymbol{q}),
\end{aligned}$$

where $\boldsymbol{\Pi}$ is a mass-weighted projector onto the tangent space of the constraints at the point $\boldsymbol{q}$ defined by

$$\boldsymbol{\Pi} = I - \boldsymbol{g}'(\boldsymbol{q})^T\left[\boldsymbol{g}'(\boldsymbol{q})\boldsymbol{M}^{-1}\boldsymbol{g}'(\boldsymbol{q})^T\right]^{-1}\boldsymbol{g}'(\boldsymbol{q})\boldsymbol{M}^{-1}.$$